COMPUTER NETWORKS FINAL EXAMINATION (2022/2023) – Moad B

Gadi Taubenfeld

Question 1

In virtual circuits – True
In the distributed algorithm – True
A user gets the IP address – True
The destination MAC address – False
When an ARP query arrives – True
Hubs do not use – True
ARP queries are always – True
In IPv6, the size – True
All the computers in the same – False
Some operating systems – True

Question 2

Consider the consensus algorithm – 10
Alice and bob applications – Address-restricted-cone, but not if it is port-restricted-cone
To get around the problem – between 10 day and 30 days
We wish to send a file – 67.5

Question 3

3.1 90 bits
3.2 $2^{28}/100$, which is about 2,684,354 bps
3.3

Section	Answer
a	**20 (not 32)**
b	**34**
c	**24**
d	**24**

Question 4

4.1 Three entries
(1) 155.128.244.0/23 outgoing link number 1,
(2) 155.128.245.0/25 outgoing link number 2,
(3) 155.128.244.128/29 outgoing link number 2.
[if somebody writes (1) 155.128.**245**.0/23 it is also ok because the mask is /23]

4.2

	Answer for question #1	Answer for question #2
Point	**Network with four routers Number of packets**	**Network with four bridges Number of packets**
(1)	**10**	**18**
(2)	**8**	**14**
(3)	**6**	**10**
(4)	**6**	**10**
(5)	**0**	**6**
(6)	**0**	**6**
(7)	**6**	**10**

	Answer for question #3	Answer for question #4
Point	**Network with router + 3 bridges Number of packets**	**Network with router + 3 hubs Number of packets**
(1)	**14**	**14**
(2)	**10**	**14**
(3)	**6**	**6**
(4)	**6**	**6**
(5)	**2**	**6**
(6)	**0**	**0**
(7)	**8**	**14**